VERITAS UNIVERSITY, ABUJA

DEPARTMENT OF SOFTWARE ENGINEERING · VUA-SEN307 DATABASE SYSTEMS

WEEK 4 PRACTICAL — LAB 2: DATABASE DESIGN AND NORMALIZATION

Scenario: A university course-registration system covering Departments, Students, Lecturers, Courses and Enrolments. The design also records which lecturer teaches which course in a particular session.

1. Unnormalized Form (UNF)

The supplied registration sheet places multiple course codes, titles, lecturers and lecturer departments in the same cells. This makes individual values difficult to query and causes repeated data and update anomalies.

matric_no	student_name	student_dept	course_codes	course_titles	lecturers	lecturer_depts	session
20231234	Ada Obi	Software Engineering	SEN307, SEN301	Database Systems, Software Architecture	Dr A. Okafor, Dr M. Bello	Software Eng., Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307, COS301	Database Systems, Operating Systems	Dr A. Okafor, Mrs F. Eze	Software Eng., Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

2. First Normal Form (1NF)

Rule: each cell holds one value and each row represents one student–course pair. The composite key for this example is (matric_no, course_code, session), assuming a student can register for a given course once per session.

matric_no	student_name	student_dept	course_code	course_title	lecturer_name	lecturer_dept	session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231235	John Musa	Computer Science	COS301	Operating Systems	Mrs F. Eze	Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

What was fixed: Multi-valued cells were split into individual rows, with one row for each student–course registration.

3. Second Normal Form (2NF)

Rule: the table is in 1NF and each non-key attribute depends on the whole candidate key. In the 1NF table, student_name and student_dept depend only on matric_no, while course_title depends only on course_code. These partial dependencies are separated into Student, Course and Enrolment.

Student

matric_no (PK)	student_name	student_dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

Course

course_code (PK)	course_title
SEN307	Database Systems
SEN301	Software Architecture

course_code (PK)	course_title
COS301	Operating Systems

Enrolment

matric_no (PK, FK)	course_code (PK, FK)	session (PK)
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027
20231235	SEN307	2026/2027
20231235	COS301	2026/2027
20231236	SEN301	2026/2027

What was fixed: Student and course descriptions were moved into their own tables; Enrolment records the student, course and session together.

4. Third Normal Form (3NF)

Rule: the table is in 2NF and non-key attributes do not depend transitively on a key. Department names are stored once in Department. Lecturer details are stored in Lecturer, and CourseTeaching records lecturer–course–session assignments. Course department assignments shown below are reasonable examples because the handout does not specify every course's department.

Table	Columns and keys
Department	dept_id (PK), dept_name (UNIQUE)
Student	student_id (PK), matric_no (UNIQUE), full_name, dept_id (FK → Department.dept_id)
Lecturer	lecturer_id (PK), lecturer_name, dept_id (FK → Department.dept_id)
Course	course_id (PK), course_code (UNIQUE), course_title, dept_id (FK → Department.dept_id)
Enrolment	student_id (PK, FK → Student), course_id (PK, FK → Course), session (PK). Composite PK: (student_id, course_id, session)
CourseTeaching	course_id (PK, FK → Course), lecturer_id (PK, FK → Lecturer), session (PK). Composite PK: (course_id, lecturer_id, session)

What was fixed: Department details are stored separately, and CourseTeaching explicitly represents the many-to-many relationship between lecturers and courses across sessions.

5. Relationship and Cardinality Summary

Relationship	Cardinality	Implementation
Department → Student	1:N	Student.dept_id FK
Department → Lecturer	1:N	Lecturer.dept_id FK
Department → Course	1:N	Course.dept_id FK
Student → Course	M:N	Resolved by Enrolment
Lecturer → Course	M:N	Resolved by CourseTeaching; session is recorded

6. Conclusion

Normalization from UNF to 1NF, 2NF and 3NF reduces repeated data and common insertion, update and deletion anomalies. The final schema supports university departments, students, lecturers, courses, enrolments and lecturer teaching assignments using primary and foreign keys.

Prepared from the VUA-SEN307 Week 4 Practical Handout. Example assumptions are identified where the handout does not specify all details.